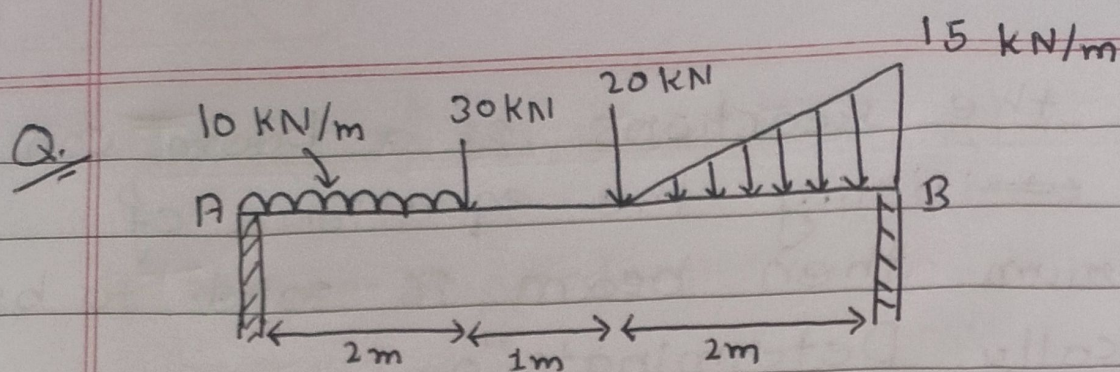
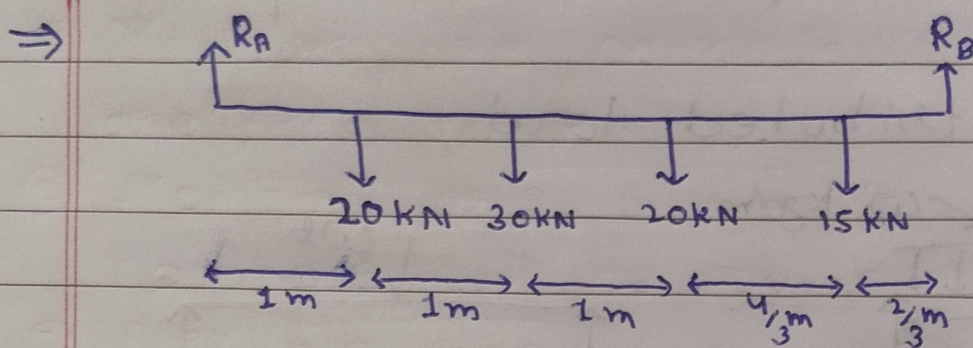




Find out the support reactions (R_A , R_B) of simply supported beam shown above.



$$\sum F_y = 0$$

$$R_A + R_B = 20 + 30 + 20 + 15$$

$$R_A + R_B = 85 \text{ kN} \quad \text{--- (1)}$$

$$\sum M_A = 0$$

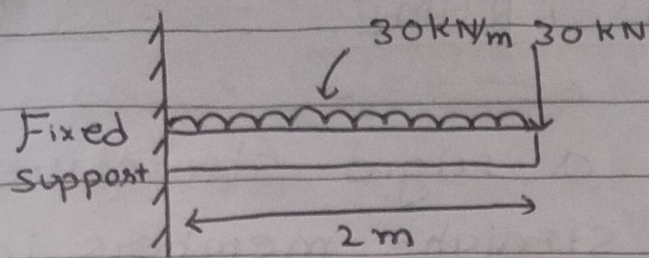
$$20 \times 1 + 30 \times 2 + 20 \times 3 + 15 \times 4.33 = R_B \times 5 \quad \text{--- (2)}$$

$$20 + 60 + 60 + 15 \times \frac{13}{3} = R_B \times 5$$

$$R_B = 41 \text{ kN}$$

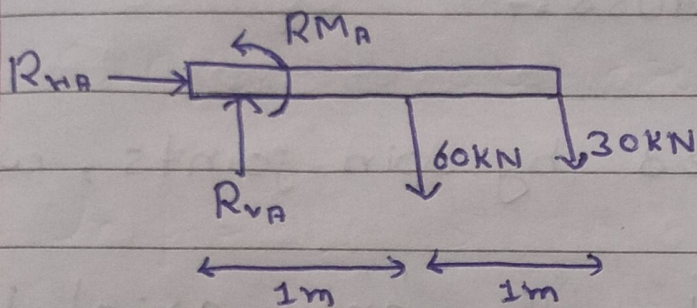
$$R_A = 85 - 41 = 44 \text{ kN}$$

Q.



Find the support reaction and moment at fixed end.

⇒



$$\sum F_x = 0$$

$$R_{HA} = 0$$

$$\sum F_y = 0$$

$$R_{VA} = 60 + 30$$

$$R_{VA} = 90 \text{ kN.}$$

$$\sum M_A = 0$$

$$-R_{MA} + 60 + 30 \times 2 = 0$$

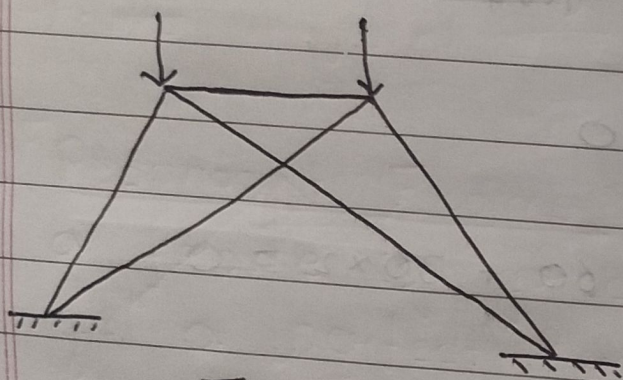
$$R_{MA} = 120 \text{ kN}\cdot\text{m}$$

TRUSS / PIN-JOINTED FRAME

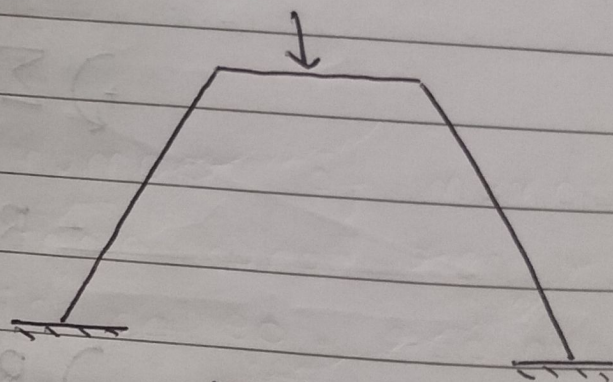
Truss is essentially a rigid structure made up of collection of straight members joined with each other through Pin joints.

Assumption for a truss:-

- ① Members are connected by pin joints, which are free to rotate.
- ② Members are subjected to only axial loads, so there will be no bending moments.
- ③ Negligible weight of members.



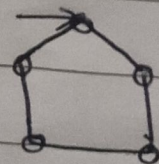
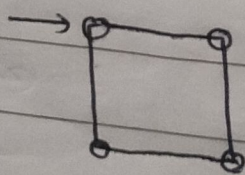
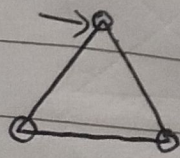
Truss

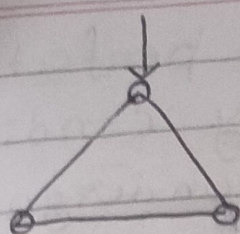


Frame

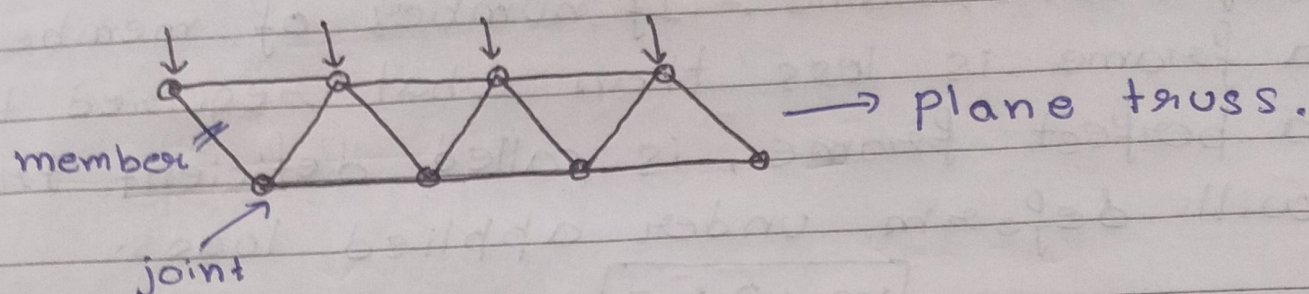
Loads are acting on pin joints only, so they can carry axial load only.

No joints are there, and bending moments are present in frame frames.





Triangular truss is the simplest perfect structure, which is stable and larger structures can be build up using triangular truss.

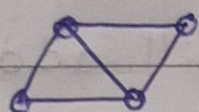


- ① Perfect frame :- If a frame has got sufficient number of members, so that it can resist the applied loads, called perfect frame. i.e. it will not undergo any deformation. necessary ~~joint~~ condition:

$$m = 2j - 3$$

m = number of members

j = no. of joints

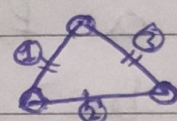


$$m = 5$$

$$j = 4$$

$$5 = 4 \times 2 - 3$$

$$5 = 5 \checkmark$$

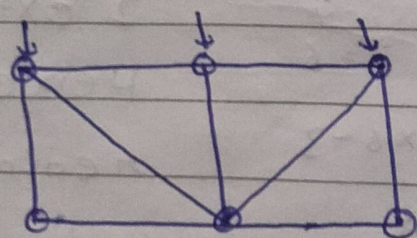


$$m = 3$$

$$j = 3$$

$$3 = 3 \times 2 - 3$$

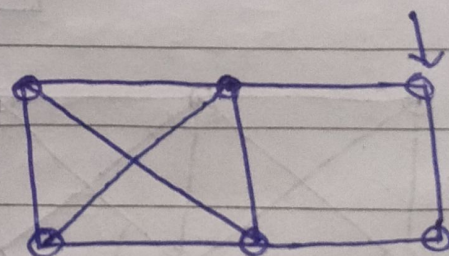
$$3 = 3 \checkmark$$



$$9 = 2 \times 6 - 3$$

$$9 = 9$$

Perfect frames.



$$9 = 2 \times 6 - 3$$

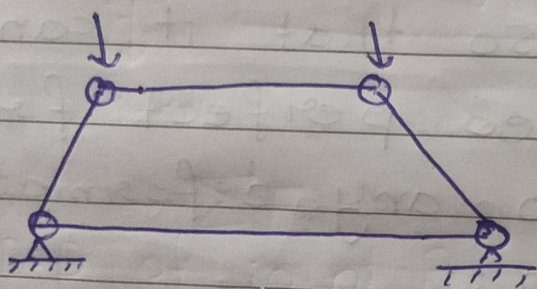
$$9 = 9$$

not a perfect frame
although satisfies
 $m = 2j - 3$

The sufficient condition for a perfect frame is that it satisfies necessary condition ($m = 2j - 3$), as well as not undergo trans under applied load.

- ② Deficient Frame :- If number of members in a frame is less than what required to be a perfect frame, is called deficient frame. It will deform under applied loads.

$$m < 2j - 3$$



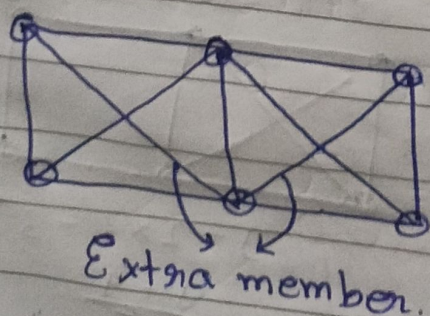
$$m = 4, j = 4$$

$$4 < 2 \times 4 - 3$$

$$4 < 5$$

- ③ Redundant Frame :- If number of members in a frame is greater than what required to be a perfect frame is called, redundant degree of Indeterminacy. So, it is also called statically Indeterminate frame.

$$m > 2j - 3$$



$$m = 11, j = 6$$

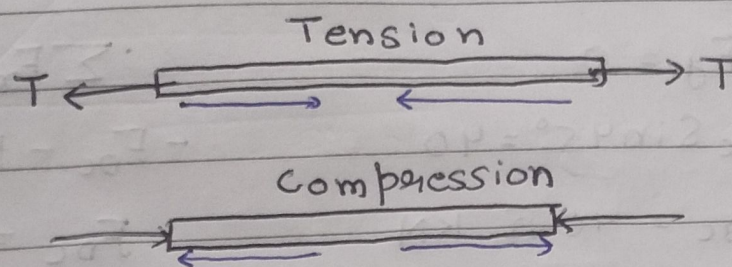
$$11 > 2 \times 6 - 3$$

$$11 > 9$$

2 extra members

Hence, 2 degree redundant frame

Nature of forces in member:-

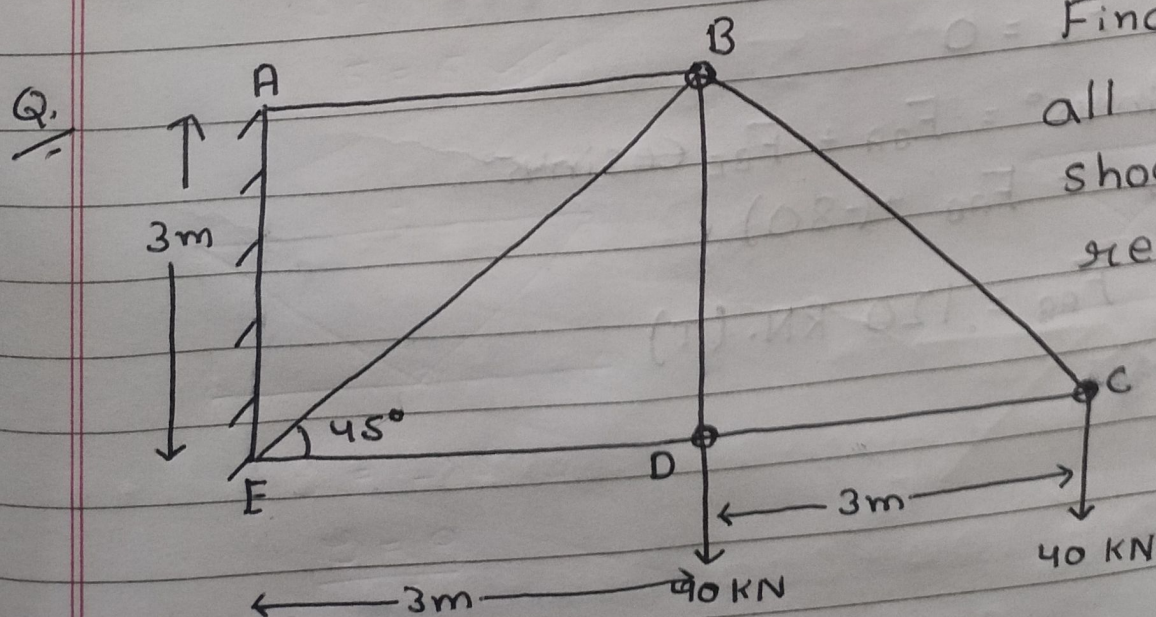


Method of Analysis:

- ① Method of joint
- ② Method of section.

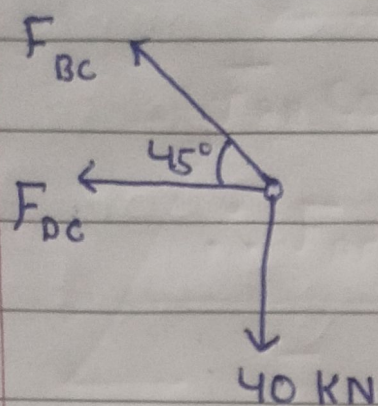
Truss Analysis (Joint Method).

1. Look for a joint where there are only two unknown are present. If such a joint is not there, then number of unknown may reduced to two at joint.
2. At the joint, the system of forces of concurrent force system then we can apply the condition of equilibrium ($\sum F_x = 0$, $\sum F_y = 0$) at the joint and find the forces in unknown member.



Find the forces in all members of truss shown. Tabulate the results.

⇒ F.B.D of joint C.



$$\sum F_y = 0$$

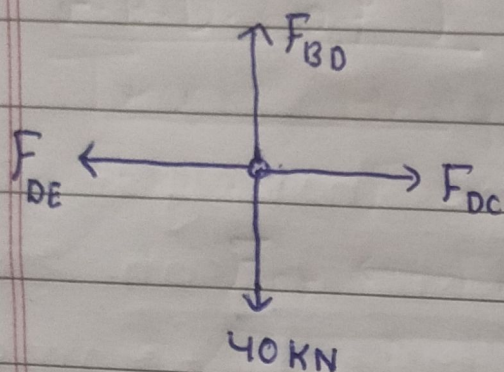
$$F_{BC} \sin 45^\circ = 40$$

$$F_{BC} = 40\sqrt{2} \text{ kN} \\ = 56.5 \text{ kN (T)}$$

$$\sum F_x = 0$$

$$-F_{DC} - F_{BC} \cos 45^\circ = 0 \\ F_{DC} = -40 \text{ kN (C)}$$

F.B.D of joint D



$$\sum F_y = 0$$

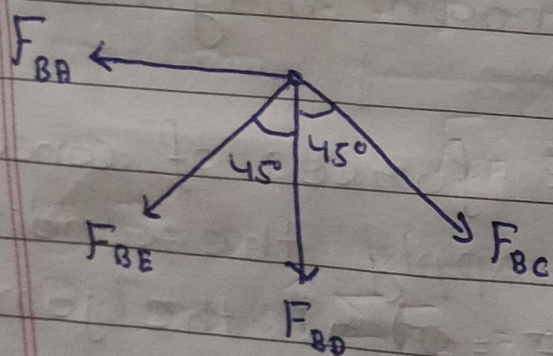
$$F_{BD} = 40 \text{ kN (T)}$$

$$\sum F_x = 0$$

$$F_{DE} = F_{DC}$$

$$F_{DE} = -40 \text{ kN (C)}$$

F.B.D of joint B.



$$\sum F_y = 0$$

$$-F_{BD} - F_{BC} \cos 45^\circ - F_{BE} \cos 45^\circ = 0$$

$$-40 - 40 - F_{BE} \cdot \frac{1}{\sqrt{2}} = 0$$

$$F_{BE} = -80\sqrt{2} = -113 \text{ kN (C)}$$

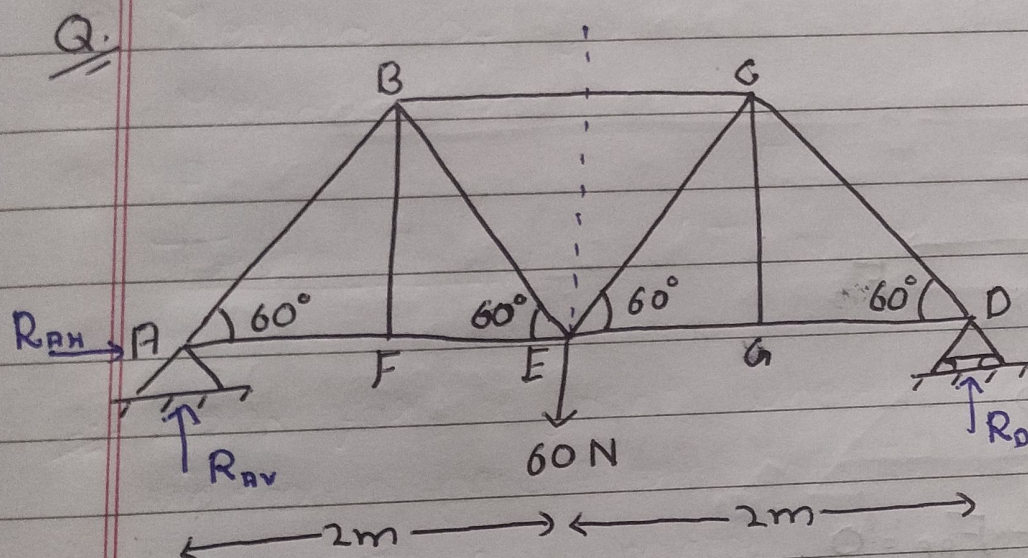
$$\sum F_x = 0$$

$$F_{BC} \sin 45^\circ = F_{BA} + F_{BE} \sin 45^\circ$$

$$40 = F_{AB} + (-80)$$

$$F_{AB} = 120 \text{ kN (T)}$$

S.No.	Force in member	magnitude	Nature
1.	F_{AB}	120	Tension
2.	F_{BC}	56.5	Tension
3.	F_{CD}	40	Compression
4.	F_{DE}	40	Compression
5.	F_{BD}	40	Tension
6.	F_{BE}	113	Compression.

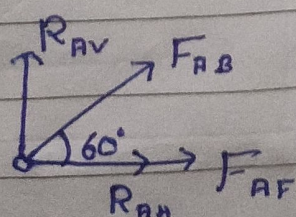


Find the force in all the members of truss and tabulate the result.

$$\begin{aligned} \rightarrow \sum F_x = 0 \quad \uparrow \sum F_y = 0 \\ R_{AH} = 0 \quad R_{Av} + R_D - 60 = 0 \Rightarrow R_{Av} = 30 \text{ kN} \end{aligned}$$

$$\begin{aligned} \uparrow \sum M_A = 0 \\ 60 \times 2 = R_D \times 4 \\ R_D = 30 \text{ kN.} \end{aligned}$$

F.B.D of joint A.



$$\begin{aligned} \uparrow \sum F_y = 0 \\ R_{Av} + F_{AB} \sin 60^\circ = 0 \\ 30 + F_{AB} \frac{\sqrt{3}}{2} = 0 \end{aligned}$$

$$\begin{aligned} F_{AB} &= -\frac{20\sqrt{3}}{1} \\ &= -34.64 \text{ kN. (C)} \end{aligned}$$

$$\rightarrow \sum F_x = 0$$

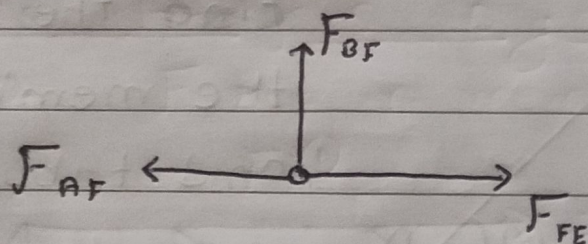
$$F_{AB} \cos 60^\circ + F_{AF} = 0$$

$$\therefore F_{AF} = -F_{AB} \cos 60^\circ$$

$$F_{AF} = -(-34.64) \cos 60^\circ$$

$$F_{AF} = 17.32 \text{ kN (T)}$$

F.B.D. of F.



$$\rightarrow \sum F_x = 0$$

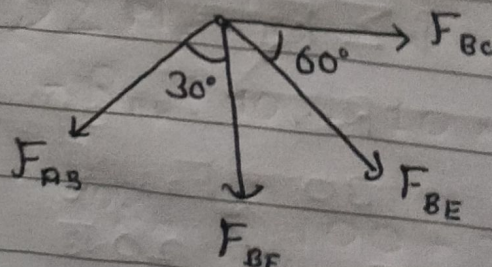
$$\therefore F_{EF} - F_{AF} = 0$$

$$F_{EF} = F_{AF} = 17.32 \text{ kN (T)}$$

$$\uparrow \sum F_y = 0$$

$$F_{BF} = 0$$

F.B.D of B.



$$\uparrow \sum F_y = 0$$

$$- F_{AB} \cos 30^\circ - F_{BF} - F_{BE} \sin 60^\circ = 0$$

$$F_{BE} \frac{\sin 60^\circ}{\cos 60^\circ} = - F_{BF} - F_{AB} \cos 30^\circ$$

$$\frac{F_{BE} \sqrt{3}}{2} = 0 + 20\sqrt{3} \cdot \frac{\sqrt{3}}{2}$$

$$F_{BE} = 60 \text{ KN (C)} \quad F_{BE} = 34.64 \text{ KN (T)}$$

$$\rightarrow \sum F_x = 0$$

$$F_{BC} + F_{BE} \cos 60^\circ = F_{AB} \sin 30^\circ$$

$$F_{BC} - 60 \times \frac{\sqrt{3}}{2} = - \frac{20\sqrt{3}}{2} \times \sqrt{3}$$

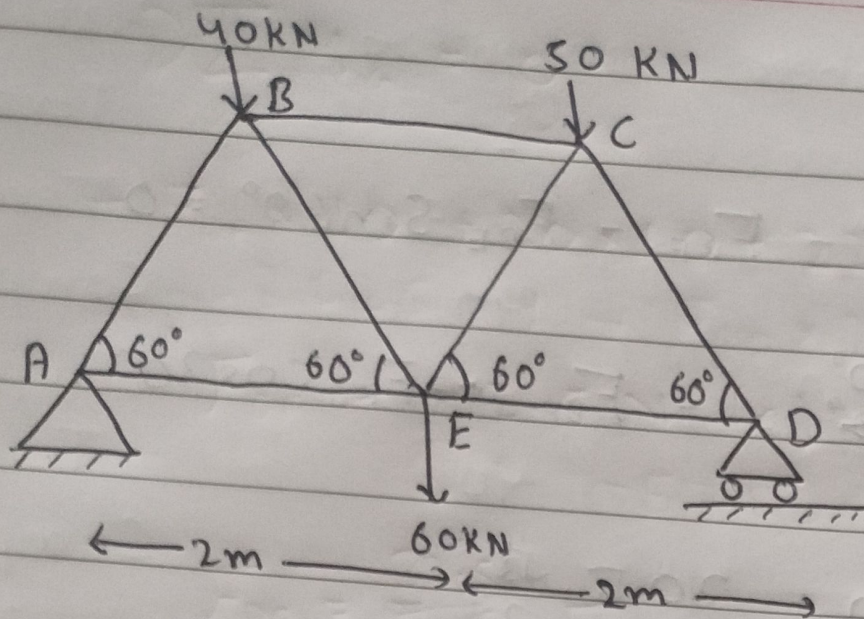
$$F_{BC} = 40\sqrt{3} = 69.28 \text{ KN (T)}$$

By symmetry of structure:

Table

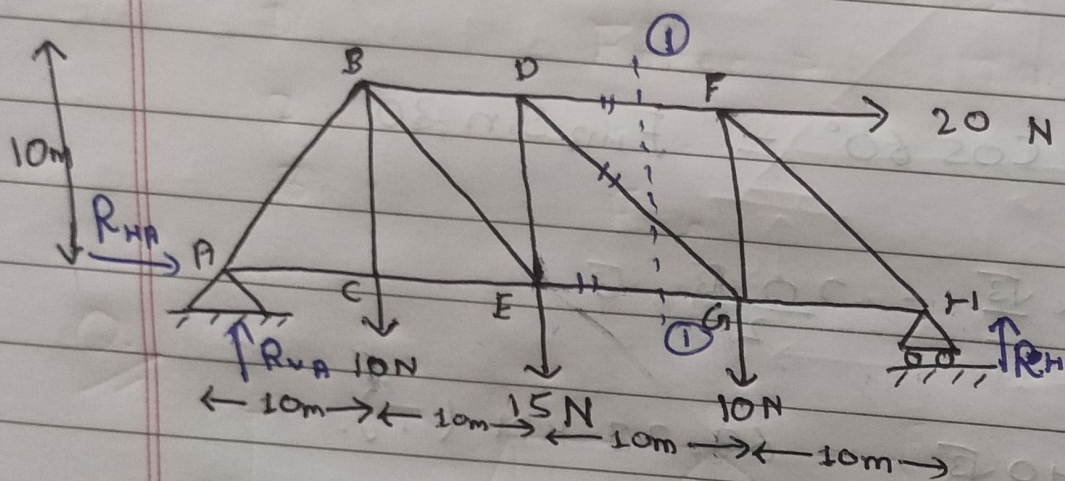
$F_{AB} = F_{CD}$	Member	Magnitude	Nature
$F_{AF} = F_{GD}$			
$F_{FE} = F_{GE}$	F_{AB}, F_{CD}	34.64 KN	C
$F_{FB} = F_{GC}$	F_{AF}, F_{GD}	17.32 KN	T
$F_{BE} = F_{CE}$	F_{FE}, F_{GE}	17.32 KN	T
	F_{FB}, F_{GC}	0 KN	-
	F_{BE}, F_{CE}	34.64 KN	T
	F_{BC}	34.64 KN	C

Q.



Determine forces all members of truss and tabulate the results.

Truss Analysis by section method.



Find the force in members DF, DG, EG

- ① Find the support rxn at A and H.
- ② Draw the sectional plane cutting not more than 3 unknown.

$$\rightarrow \sum F_x = 0$$

$$R_{HA} = -20 \text{ N}$$

$$\uparrow \sum F_y = 0$$

$$R_{VA} + R_H - 35 = 0$$

$$R_{VA} + R_H = 35 \text{ N}$$

$$\curvearrowright \sum M_A = 0$$

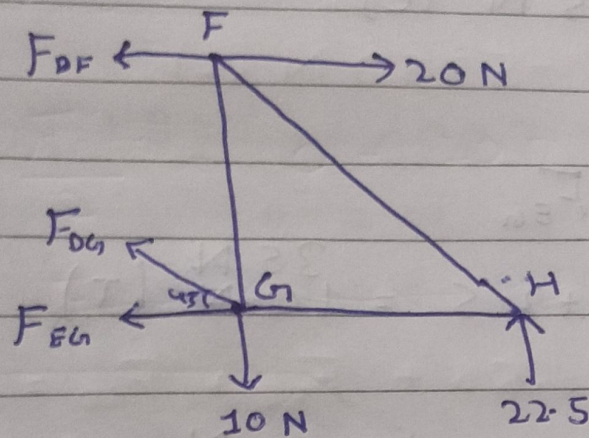
$$-R_H \times 40 + 10 \times 10 + 15 \times 20 + 10 \times 30 + 20 \times 10 = 0$$

$$40R_H = 100 + 300 + 300 + 200$$

$$R_H = \frac{900}{40} = 22.5 \text{ N}$$

$$R_{VA} = 35 - 22.5 \\ = 12.5 \text{ N.}$$

F.B.D of R.H.S part of section ①-①



Apply Equilibrium conditions

$$\Rightarrow \sum F_x = 0 \quad \uparrow \sum F_y = 0 \quad \Rightarrow \sum M_G = 0$$

$$\uparrow \sum F_y = 0$$

$$F_{DG} \sin 45^\circ - 10 + 22.5 = 0$$

$$F_{DG} \times \frac{1}{\sqrt{2}} = -12.5$$

$$F_{DG} = -12.5\sqrt{2} = -17.67 \text{ N (C)}$$

$$\Rightarrow \sum F_x = 0$$

$$-F_{DF} + 20 \text{ N} - F_{DG} \cos 45^\circ - F_{EG} = 0$$

$$F_{DF} = 20 + 12.5 - F_{EG}$$

$$F_{DF} = 7.5 - F_{EG} \quad 32.5 - F_{EG}$$

$$\sum M_G = 0$$

$$-F_{DF} \times 10 + 20 \times 10 - 22.5 \times 10 = 0$$

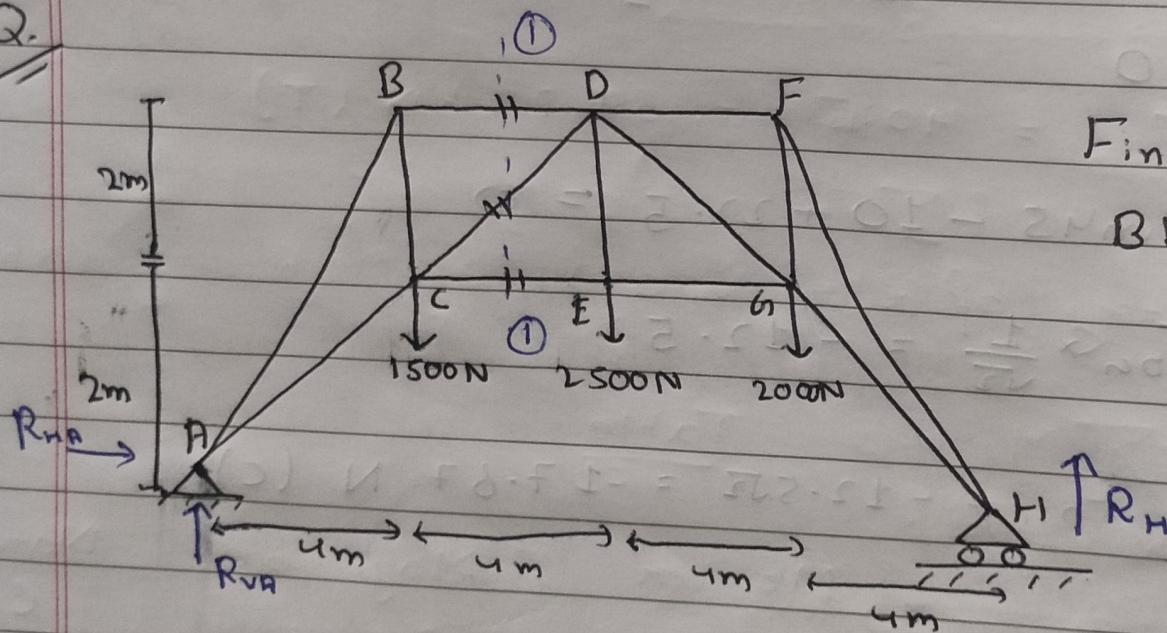
$$F_{DF} = -2.5 \text{ N (C)}$$

$$F_{DF} = \frac{32.5}{7.5} - F_{EG}$$

$$F_{EG} = \frac{32.5}{7.5} + 2.5 = 10 \text{ N (T)}$$

S. No.	Force in member	Magnitude	Nature
1	F_{DF}	2.5 N	C
2	F_{DG}	17.67 N	C
3	F_{EG}	35 N	T

Q.



Find force in member
BD, CD, CE.

$$\sum F_x = 0$$

$$R_{HA} = 0$$

$$\sum F_y = 0$$

$$R_{VA} + R_H - 1500 - 2500 - 2000 = 0$$

$$R_{VA} + R_H = 6000 \text{ N}$$

$$\sum M_A = 0$$

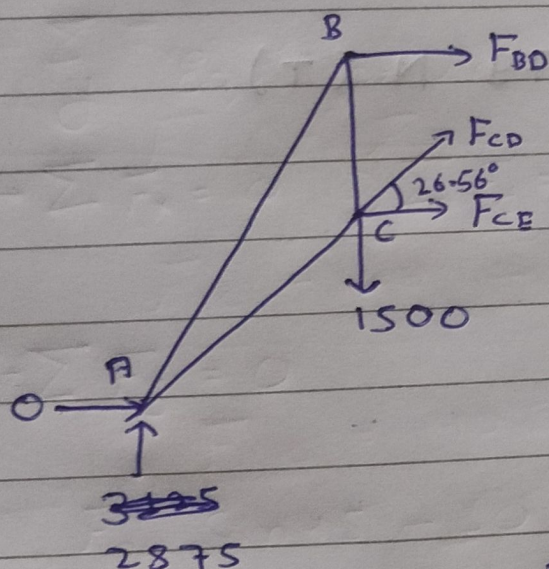
$$1500 \times 4 + 2500 \times 8 + 2000 \times 12 - R_H \times 16 = 0$$

$$R_H = 3125 \text{ N}$$

$$R_{VA} + 3125 = 6000 \text{ N}$$

$$R_{VA} = 2875 \text{ N}$$

F.B.D of L.H.S part of section ①-①



$$\sum F_x = 0$$

$$F_{BD} + F_{CE} = 0$$

$$\sum M_C = 0$$

$$4 \times 3125 + 2 \times F_{BD} = 0$$

$$F_{BD} = -6250 \text{ N (C)}$$

$$\sum F_x = 0$$

$$F_{CE} + F_{CD} \cos 26.56 = 0$$

$$\sum F_y = 0$$

$$2875 - 1500 + F_{CD} \sin 26.56 = 0$$

$$F_{CD} = -229.69 \text{ N (C)}$$

$$F_{CD} = -479.69 \text{ N (C)}$$

$$\sum M_c = 0$$

$$2875 \times 4 + F_{BD} \times 2 = 0$$

$$F_{BD} = -5750 \text{ N (c)}$$

$$\sum F_x = 0$$

$$F_{CD} \cos 26.56 + F_{CE} + F_{BD} = 0$$

$$F_{CE} = 6179 \text{ N (T)}$$